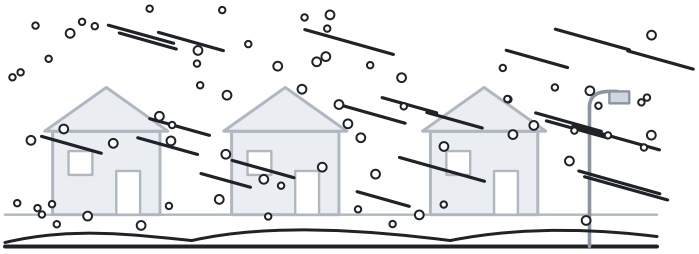
Storm Chasers: Practice Test 1

Time: 50 minutes. Work with your partner at 14 stations, about 3 1/2 minutes each. When the coach calls 'Rotate!', move to the next station; you may not go back, even if a station is empty. Bring only pencils: no notes, flashcards, or books, just like the tournament.

Partner names: _____ and _____ Score: _____

Station 1: Whiteout

Look at the picture.



1. What kind of severe weather is this?

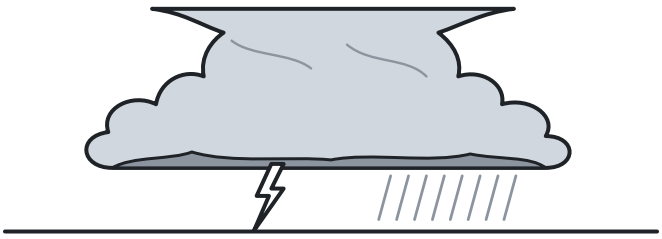
◯ A Hurricane ◯ B Flood ◯ C Blizzard ◯ D Tornado

2. Name the 3 things that make a snowstorm a blizzard.

3. Name 2 ways to stay safe during a blizzard.

Station 2: Building a thunderstorm

Look at the picture of the storm cloud.



4. What is the name of this tall, dark thunderstorm cloud?

5. Name the 3 ingredients a thunderstorm needs.

6. All severe weather comes from changes in Earth's _____.

7. Name 3 hazards a severe thunderstorm can bring.

Station 3: Flash to bang

Maya is at the park. She sees a flash of lightning and counts 30 seconds until she hears thunder.

8. How far away is the lightning? Show your math.

9. Is Maya safe from lightning if she stays outside? Explain.

10. Her friend says, 'Let's wait under that big tree.' Is that a good idea? Where should they go?

11. The last thunder is heard at 3:10 pm. What is the earliest time they should go back outside?

Station 4: Tornado damage

After a tornado, only a pile of boards is left where a house once stood.



12. Which wind speed best fits this damage?

○ A 70 mph ○ B 100 mph ○ C 130 mph ○ D Over 170 mph

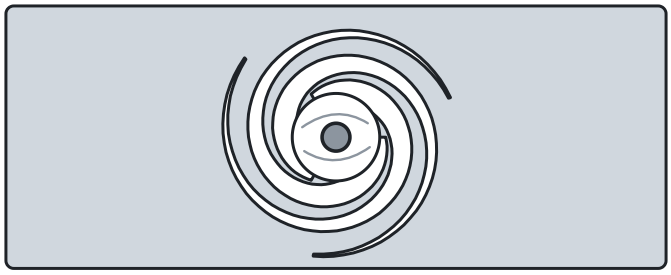
13. What does the EF scale use to rate a tornado?

14. A spinning cloud hangs down from a storm but has not touched the ground. What is it called?

15. A tornado warning is announced during school. Where should students go, and what should they do?

Station 5: Picture from space

This picture of a storm was taken from high above Earth.



16. What kind of storm is this, and what tool took the picture?

17. What is the calm center called? What is the ring of the strongest winds around it called?

18. What are the spiral arms of storms called?

19. Why does a hurricane get weaker after it moves over land?

Station 6: Storm tracker

Top wind speeds of a storm each day: Monday 30 mph. Tuesday 45 mph. Wednesday 80 mph. Thursday 110 mph. Friday (after moving over land) 50 mph.

20. What was the storm on Monday?

◯ A Tropical depression ◯ B Tropical storm ◯ C Hurricane

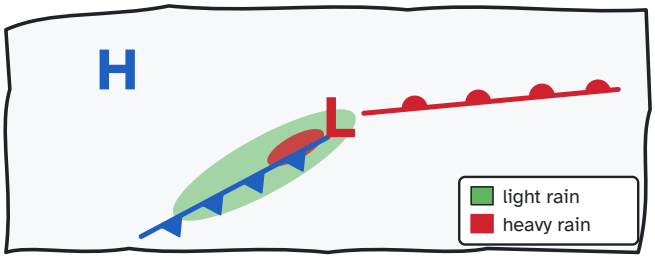
21. On which day did the storm first become a hurricane? How do you know?

22. On which day did the storm first get a name? Why?

23. What was the storm on Friday, and why did it change?

Station 7: Read the map

Use the weather map and its key.



24. What does the red line with half circles show?

25. What does the L stand for, and what weather does it usually bring?

26. Where on this map is the rain the heaviest? How can you tell?

27. The cold front moves past a town. What will the weather be like after it passes?

Station 8: Watch or warning?

Choose watch or warning for each message.

28. 'Conditions are right for tornadoes in Wake County until 9 pm.'

◯ A Watch ◯ B Warning

29. 'Flash flooding is happening now along Crabtree Creek.'

◯ A Watch ◯ B Warning

30. 'A severe thunderstorm with quarter-size hail is moving into Raleigh now.'

◯ A Watch ◯ B Warning

31. What should your family do during a tornado WATCH?

Station 9: Water on the road

Heavy rain fell for 2 hours. Water is rushing over a low road.

32. What is a flash flood?

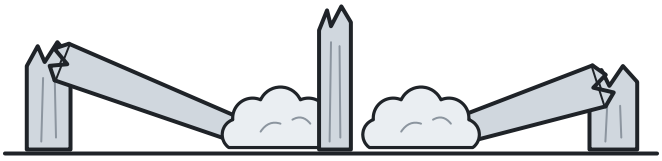
33. How deep does moving water need to be to knock a person down? To carry away a car?

34. What does 'Turn Around, Don't Drown' tell people to do?

35. Name 2 things that can cause flooding.

Station 10: After the storm

A strong storm went through a forest.



36. Name 2 kinds of severe weather that could snap trees like this.

37. Name one way this damage hurts animals and one way it can help other living things.

38. How can lightning change a forest?

39. Name one GOOD thing a river flood can leave behind on a floodplain.

Station 11: Changing the land

Two hills get the same heavy rain. Hill A is covered with grass and trees. Hill B is bare dirt.

40. Which hill loses more soil? Why?

41. The rain carves small channels down Hill B. What are they called?

42. The soil washes down and piles up at the bottom of the hill. What is this dropping of soil called?

43. Storm surge from a hurricane hits a beach. Name 2 ways the land can change.

Station 12: Weather tools

Choose the best tool for each job.

44. Measures air pressure.

◯ A Thermometer ◯ B Barometer ◯ C Rain gauge ◯ D Wind vane

45. Shows which way the wind is blowing.

◯ A Thermometer ◯ B Barometer ◯ C Rain gauge ◯ D Wind vane

46. Shows where rain and hail are right now and can spot a storm that is spinning.

◯ A Weather balloon ◯ B Anemometer ◯ C Doppler radar ◯ D Rain gauge

47. The barometer reading drops all afternoon. What might be coming? Why?

Station 13: A real NC storm

In October 2016, Hurricane Matthew passed along the North Carolina coast. It dropped more than 10 inches of rain on parts of eastern NC in about a day. Rivers kept rising for days, and towns like Lumberton and Princeville flooded.

48. What caused most of the damage in eastern NC?

49. Why did rivers keep flooding for days after the rain stopped?

50. Name 2 ways a community gets ready before a hurricane.

51. Name 3 things that belong in a family emergency kit.

Station 14: Compare the storms

52. Which usually lasts longer and covers a bigger area: a tornado or a hurricane?

53. Name one way a thunderstorm and a hurricane are alike.

54. Which severe weather is most likely in Raleigh on a hot July afternoon?

◯ A Thunderstorm ◯ B Blizzard ◯ C Hurricane ◯ D Tornado

55. Which scale is used to rate hurricanes, and which is used to rate tornadoes?

28	A Watch.
29	B Warning.
30	B Warning.
31	Get ready: know your safe place, have your kit, and keep listening for a warning.
Station 9: Water on the road	
32	A flood that happens very fast, in minutes to a few hours, usually after heavy rain.
33	About 6 inches can knock you down; about 12 inches can carry away a car.
34	Never walk or drive through flood water. Turn around and find another way.
35	Any 2: heavy rain, a slow storm that rains for days, melting snow, storm surge, rivers overflowing, a broken dam.
Station 10: After the storm	
36	Any 2: tornado, hurricane, severe thunderstorm (strong straight-line winds). Also accept ice storm or blizzard with a reason.
37	Hurts: nests and homes are knocked down. Helps: fallen logs become homes for insects and fungi, and more sunlight reaches the ground so new plants grow.
38	It can start wildfires and split or kill trees. (Some pine forests need fire to stay healthy.)
39	New, rich soil that helps plants grow.
Station 11: Changing the land	
40	Hill B. It has no roots to hold the soil in place.
41	Gullies.
42	Deposition.
43	Any 2: sand dunes wash away, the beach gets narrower, sand is pushed inland, new inlets are cut through islands, salt water floods marshes.
Station 12: Weather tools	
44	B Barometer.
45	D Wind vane.
46	C Doppler radar.
47	A storm. Falling air pressure means low pressure is moving in, which often brings clouds, rain, and storms.
Station 13: A real NC storm	
48	Flooding from heavy rain (rivers overflowing).
49	Rain water from all the land upstream kept draining into the rivers and flowing downstream, so the water kept rising.
50	Any 2: sends alerts, opens shelters, tells people to evacuate, fills sandbags, clears storm drains, gets rescue crews ready.
51	Any 3: water, food that will not spoil, a can opener, flashlight and batteries, first aid kit, medicine, weather radio, phone charger, blankets.
Station 14: Compare the storms	
52	A hurricane (days long, hundreds of miles wide). A tornado lasts minutes and is usually less than 1 mile wide.
53	Any 1: both form when warm, wet air rises; both bring heavy rain, strong winds, and lightning; both can cause floods and tornadoes.
54	A Thunderstorm.

55	Hurricanes: Category 1 to 5 (the Saffir-Simpson scale). Tornadoes: the EF scale (EF0 to EF5).